

Assignment 7

WSS and SSS

1. A random process is given by $X(t) = \cos \omega t$, where ω is a random variable uniform in $(0, 1)$. Prove that $\{X(t)\}$ is not stationary.
2. Show that the random process $X(t) = A \cos(\omega_0 t + \Theta)$ where A and ω_0 are constants and Θ is a uniform random variable in $(0, \pi)$ is not stationary.
3. A random process is given by $X(t) = A \cos(\omega_0 t + \Theta)$ where ω_0 is a constant and Θ is a random variable uniform in $(0, \pi/2)$. Show that $\{X(t)\}$ is not wide sense stationary.
4. A random process is given by $X(t) = A \cos \omega_0 t$ where ω_0 is a constant and A is uniformly distributed over $(0, 1)$. Show that $\{X(t)\}$ is not stationary.
5. Show that the random process $X(t) = 10 \cos(100t + \Theta)$ where Θ is a random variable uniformly distributed on $(-\pi, \pi)$ is a wide-sense stationary process.
6. Show that the random process $X(t) = \sin(\omega_0 t + Y)$ where ω_0 is a constant and Y is a random variable uniformly distributed over $(0, 2\pi)$ is wide-sense stationary.
7. A random process is defined by $X(t) = A \sin(\omega_0 t + \Theta)$ where A and ω_0 are constants and Θ is a random variable uniformly distributed over $(-\pi, \pi)$. Is $X(t)$ stationary?
8. A random process is defined by $X(t) = \sin(\omega_0 t + \Theta)$ where Θ is uniformly distributed in $(0, 2\pi)$ and ω_0 is constant. Is $\{X(t)\}$ wide-sense stationary?
9. Show that the random process given by $X(t) = A e^{(\omega_0 t + \Theta)}$ is stationary in wide-sense if $E(A) = 0$ and ω_0, Θ are constants.
10. A WSS random process $\{X(t)\}$ has autocorrelation $R_{X,X}(\tau) = A e^{-\alpha|\tau|}$. Find the second moment of the random variable $Y = X(7) - X(4)$.
11. If $X(t) = X_1(t) + t \cdot X_2(t)$ where X_1 and X_2 are independent random variables with $E(X_1) = a$, $E(X_2) = b$, $\text{Var}(X_1) = \sigma_1^2$, $\text{Var}(X_2) = \sigma_2^2$, find $E[X(t)]$, $R_X(t_1, t_2)$, $C_X(t_1, t_2)$. Is the process stationary?
12. If $X(t) = \sin(\omega t + Y)$ where Y is uniformly distributed in $(-\pi, \pi)$, check whether $X(t)$ is WSS.
13. If $X(t) = A \cos t + B \sin t$ where A and B are independent discrete random variables each of which assumes values -1 and 1 with probabilities $1/2$ and $1/2$ respectively, prove that $X(t)$ is wide-sense stationary.
14. If $X(t) = A \cos 2\pi t$ where A is some random variable, check whether $X(t)$ is stationary.
15. If $X(t) = X_1(t) + X_2(t)$ where $X_1(t)$ and $X_2(t)$ are independent random variables uniformly distributed in $(0, 2)$ and $(0, 3)$ respectively, find $E[X(t)]$, $R_{X,X}(t_1, t_2)$, $C_{X,X}(t_1, t_2)$. Is the process stationary?